

Course Delivery Flow

Notes on Writing and Saving Your First File

For the writing and saving your first file slide:

- Ensure the students create and save a .py and .ipynb file.
- Demonstrate the process.
- Saving each file should trigger the extension installations in VS Code.

On reaching String Challenges I - if the .ipynb extension is not installed the magic command %pip install ipykernel should help.

If VS Code still looks like it's spinning or frozen:

1. Press Ctrl + Shift + P (Windows) or Cmd + Shift + P (Mac).
2. Type **Developer: Reload Window** and press Enter.
3. Try running the cell again. This clears any cached, broken connection attempts.

Command Line

Slide 91 - I'm very much used to Linux - using the Windows Powershell is often very different so forgive me if I look like I don't know what I'm talking about.

Slide 92 - Once we've made the folder, navigate into it.

- Then cd .. out of it
- Delete the folder using
 - rm folder_name (Linux)
 - rm -rf folder_name (Linux force)
 - rm -recurse - force folder_name (Powershell)

No challenges - preparatory knowledge only.

Git and GitHub

⚠ Do not attempt the steps as we work through the slides.

We will go through them together at the end

Slide 100 - Clarify that GitHub isn't the only option but it's just what I use so have chosen to use it for the course.

- GitLab
- BitBucket
- ...others?

The processes for set up are very similar, just different URLs, different accounts etc.

Slide 102 Clone Your First Repo - follow these steps.

- In the GitHub UI, create a new repo and copy the SSH URL.
- Return to VS Code. In terminal ensure you're in the folder in which you have your Workbook 1.
- Clone the repo.
- Save and close Workbook 1.
- Using the VS Code Explorer side bar, drag Workbook 1 into the cloned repo.
- Demo how to do this in terminal
 - From the folder containing the file and the repo
 - Using the CLI

```
mv workbook_1.ipynb repo_name
```

```
cp workbook_1.ipynb repo_name
```

- Demonstrate `git status` - shows the difference between the local and remote versions of the repo

```
git add .
git commit -m "my first commit"
git push origin main
```

- run `git status` again to see that the versions match
- Go to the UI and look at the changes
- Mention `git pull` for matching remote changes

Good Habits

Get into the habit of pushing your changes - each time you've made significant changes, or even just at the end of the day - assuming you're the only one likely to be working on the repo - ctrl s and go through the `add`, `commit`, `push` processes.

UV and VENV

⚠ Be clear about what the package installers do and go through the installation before moving on to VENV

Once we've done the installation together, move onto the VENV slides, using the pop out on the UV and PIP slide as a warning as to why we use VENVs and why they're important.

- Version clashes

Slide 4 - `uv init`

Go back to the terminal and do this for the repo we've just cloned.

- Make sure you're in the repo you cloned from github at its upmost level
- For good practise, check it's status
 - run `git status`
- If remote and local match, run `uv init`
- Talk about specific versions of Python

- Activate the venv if required
- Add to the .gitignore
 - using the commands on the slides
 - or editing in VS Code
- Push the repo to GitHub

I also use a VM as another layer of protection and also because it lets me operate in Linux on the terminal from my windows environment by SSH-ing into a Linux VM from VS Code

⚠ Show how to connect and leave it at that!

Data Ingestion

⚠ That was a massive segue away from looking at core Python and data, but it's important stuff.

Open METEO

🚩 [Free Open-Source Weather API | Open-Meteo.com](#)

Notebook in repo.

REST - Countries

Once you have an account

- API keys URL [API Keys - REST Countries](#)

Full text search example:

```
import requests
response = requests.get(
    'https://api.restcountries.com/countries/v5?q=stan&limit=5',
    headers={'Authorization': 'Bearer rc_live_1c4aea7fef564b5c88afed8bd009095'}
)
data = response.json()
```